

Artin 2

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Problem 4.2.

Proof. Let G be a nonabelian group with order 55. Thus $|G| = 5 * 11$, by the Sylow theorems there is a sylow subgroup of order 5 and 11. Let n be the number of sylow 11 subgroups, $n|5$ and $n \mod 11 = 1$ thus $n = 1$. Thus the Sylow 11 subgroup is normal in G . The number of sylow 5 subgroups m , $m|11$ and $m \mod 5 = 1$ thus $m = 1$ or $m = 11$. The sylow 11, is $N = \mu_{11}$ and the sylow 5 subgroup is $H = \mu_5$. $\mu_{11} \cap \mu_5 = e$ because μ_{11} and μ_5 have no proper nontrivial subgroups. Suppose μ_5 is normal. Let $\forall h \in \mu_{11}, \forall k \in \mu_5$, thus $hkh^{-1} \in \mu_5$ and $khk^{-1} \in \mu_{11}$ thus $hkh^{-1}k^{-1} \in \mu_5 \cap \mu_{11}$, thus $hkh^{-1}k^{-1} = e \implies hk = kh$ and G is abelian. Thus by contradiction μ_5 must not be normal.

Thus there are 11 sylow subgroups of order 5.

$\phi : \mu_5 \rightarrow \text{Aut}(\mu_{11}) = \mu_{10}$, thus $x \rightarrow x^2$ or $x \rightarrow e$ to maintain the relations thus $G = \mu_{11} \rtimes \mu_5 = \langle x, y | x^{11} = y^5 = e, yxy^{-1} = x^2 \rangle$. Thus $xyx^{-1} = xyx^{-1}y^{-1}y = xx^9y = x^{10}y$. Thus there are 4 more conjugation classes, $[y], [y^2], [y^3], [y^4]$. A normal subgroup is a union of conjugacy classes. So 10 breaks up, however the sizes of the conjugacy classes divides the group because $|G| = Z(g)|[g]|$, by orbit stabilizer. Thus $10 = 5 + 5$, or 2 conjugacy classes of order 5.

Thus the class equation is $1 + 5 + 5 + 11 + 11 + 11 + 11$. Thus there are 7 conjugation classes thus 6 irreducibles.

By the previous homework, $55 = \sum_{i=1}^7 (\dim V_i)^2$ where V_i is an irreducible. The dimensions of the irreducibles must divide G . Thus the only possible dimensions are 1, 5, 11. $G/\mu_{11} \cong \mu_5$ is abelian. $xyx^{-1}y^{-1} = xx^9 = x^{10}$ which generates μ_{11} . Thus $\mu_{11} = [G, G]$. Thus $[G : \mu_{11}] = 5$. Thus there are 5, 1 dimensional irreps of G . Thus the only possible combination which works is 5 1 dimensional and 2 5 dimensional irreducibles. $\square$

Problem 4.3a. Determine the character tables for K_4 group.

Proof. $K_4 = \mu_2 \times \mu_2$ is abelian so all the irreducibles are 1 dimensional and there are 4 of them because $4 = 4(1)$. $K_4 = \langle x, y | x^2 = y^2 = e, xy = yx \rangle$. Also each element is in its own conjugacy class. Both x and y can go to ± 1 . So the character table is as follows:

	e	x	y	xy
χ_1	1	1	1	1
χ_2	1	-1	1	-1
χ_3	1	1	-1	-1
χ_4	1	-1	-1	1

$\square$

Problem 4.3b. the quaternion group

Proof. $H = \langle x, y | x^4 = y^4 = e, x^2 = y^2, yx = x^{-1}y \rangle$. $|H| = 8$. Since the dimensions of the irreps divides 8 the possible dimensions are 1, 2, 4, 8. However since $\sum_i (\dim V_i)^2 = 8$, having a irreducible of size 4 is not possible since $16 > 8$. So the possible dimensions are 1 or 2. There is the trivial 1 dimensional representation, thus there are 4 1 dimensional representation and 1 2 dimensional representations.

For the 1 dimensional representations, if you send $-1 \rightarrow 1$ then $x, y \rightarrow \pm 1$. Thus there are 4 1 dimensional representations.

For the 2 dimensional representations, if you send $x^2 = y^2 \rightarrow I$ then $x, y \rightarrow \pm I$, but then all elements of G are then mapped to scalar matrices which fix any subspace of a vectorspace, thus a G -invariant subspace gets created and the 2 dimensional representation is no longer irreducible.

If you send $-1 \rightarrow -I$. Then $x \rightarrow \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$, $x^2 = -I$. Let v be a eigenvector in the i eigenspace. $xyv = yy^{-1}xyv = yx^{-1}v = -iyv$. But $y^2v = -v$. Thus under the basis v, yv , $y \rightarrow \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. x 's eigenvectors are e_1, e_2 and y 's eigenvectors are $\begin{bmatrix} -i \\ 1 \end{bmatrix}$ and $\begin{bmatrix} i \\ 1 \end{bmatrix}$. Thus there is no g -invariant subspace. Thus this is the irreducible 2 representation of H .

By last semester, the conjugacy classes of H are the following $[1], [-1], [x], [y], [xy]$. χ , 1 - 4 corresponding to 1 dimensional representations and χ_5 correspond to the 2 dimensional representation.

	e	x^2	x	y	xy
χ_1	1	1	1	1	1
χ_2	1	1	-1	1	-1
χ_3	1	1	1	-1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-1	0	0	0

□

Problem 4.3c. the dihedral group D_4 ,

Proof. $D_4 = \langle x, y | x^4 = y^2 = e, yxy^{-1} = x^{-1} \rangle$. $|D_4| = 8$.

For the dimension 1 irreducibles, since $y^2 = e$, $y \rightarrow \pm 1$. Furthermore since $x^4 = 1$ and $yxy^{-1} = x^{-1} \implies yxy^{-1}x = e$ and in the 1 dimensional representations, which are abelian, specifically $\mathbb{C}^*$, it means $x^2 = e$. Thus $x \rightarrow \pm 1$. Thus there are 4 1 dimensional irreducibles of D_4 . Since dimensions of the irreducibles must divide 8 and the sum of the squares must add up to 8, there is only 1 irreducible of dimension 2.

Let V_1, V_2, V_3, V_4 be the 1 dimension representation of D_4 . Let V_5 be the 2 irreducible dimensional representation. Furthermore since $\chi_{reg} = \sum_{i=1} \dim V_i * \chi_{V_i}$. But $\chi_{reg}(e) = 8$ and $\chi_{reg}(g) = 0 \forall g \in D_4 \ni g \neq e$. $\chi_{V_5} = \frac{1}{2}(\chi_{reg} - \sum_{i=1}^4 \dim V_i * \chi_{V_i})$. Use this to find the characters of the 2 dimensional irreducibles, the calculated values are below. Furthermore there are 5 conjugacy classes (from last semester), e, x^2, x, y, xy .

	e	x^2	x	y	xy
χ_1	1	1	1	1	1
χ_2	1	1	-1	1	-1
χ_3	1	1	1	-1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	0	0	0
χ_{reg}	8	0	0	0	0

□

Problem 4.3d. the dihedral group D_8 ,

Proof. $D_6 = \langle x, y | x^6 = y^2 = e, yxy^{-1} = x^{-1} \rangle$. $|D_6| = 12$.

For the dimension 1 irreducibles, since $y^2 = e$, $y \rightarrow \pm 1$. Furthermore since $x^6 = 1$ and $yxy^{-1} = x^{-1} \implies yxy^{-1}x = e$ and in the 1 dimensional representations, which are abelian, specifically $\mathbb{C}^*$, it means $x^2 = e$. Thus $x \rightarrow \pm 1$. Thus there are 4 1 dimensional irreducibles of D_4 . Since dimensions of the irreducibles must divide 12 and the sum of the squares must add up to 12, there is only 2 irreducible of dimension 2. So $4 + 4 + 4 = 12$.

From the irreducible representations of D_n homework, we know the 2 dimension irreducibles of D_6 have the following form. $x \rightarrow \begin{bmatrix} \omega^k & \\ & \omega^{-k} \end{bmatrix}$ where $\omega = e^{\pi i/3}$ (6th root of unity) and $k \in \{1, 2\}$ and $y \rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. In either representation, the trace of y and xy is 0. $tr(x) = \omega^k + \omega^{-k} = 2\cos(k\frac{\pi}{3}) = (-1)^{k+1}\sqrt{3}$

	e	x^2	x	y	xy
χ_1	1	1	1	1	1
χ_2	1	1	-1	1	-1
χ_3	1	1	1	-1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	$\sqrt{3}$	0	0
χ_6	2	-2	$-\sqrt{3}$	0	0
χ_{reg}	12	0	0	0	0

□

Problem 4.3d. a nonabelian group of order 21

Proof. From the irreducible representations of dihedral group homework, $G = \langle x, y | x^7 = y^3 = 1, y^{-1}xy = x^2 \rangle$.

We know the dimension of the irreducibles must divide 21 and the sum squares of the dimensions must sum to 21. So the possible dimensions of the irreducibles are 1, 3.

$xyx^{-1} = yy^{-1}xyx^{-1} = yx$, $xyxx^{-1} = xy$, $xyx^{-1} = xyx^{-1} = xy$. Thus $[y] = \{y, xy, yx\}$.

$xy^2x^{-1} = yy^{-1}xyyx^{-1} = yx^2yx^{-1} = yxyx = yx^2y = xy^2$. Thus $[y^2] = \{y^2, xy^2, y^2x\}$

$[x] = \{x, x^2, x^4\}$ and $[x^3] = \{x^3, x^5, x^6\}$.

And $[e] = e$.

Thus there are 5 conjugacy classes. For this sum to work out the only combination which works is 3 1 dimensional conjugacy classes and 2 3 dimensional irreducibles.

From the irreducibles of the dihedral homework, we know $x \rightarrow 1$ and $y \rightarrow e^{2ik\pi/3}$ where $k = 0, 1, 2$. Let $\omega = e^{2i\pi/3}$

Let $\lambda = e^{2\pi/7}$ From the irreducibles of the dihedral homework, we know the two reps are as follows:

$$x \rightarrow \begin{bmatrix} \lambda & & \\ & \lambda^2 & \\ & & \lambda^4 \end{bmatrix}, y \rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

and

$$x \rightarrow \begin{bmatrix} \lambda^3 & & \\ & \lambda^5 & \\ & & \lambda^6 \end{bmatrix}, y \rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

	e	x	x^3	y	y^2
χ_1	1	1	1	1	1
χ_2	1	1	1	ω	$\bar{\omega}$
χ_3	1	1	1	$\bar{\omega}$	ω
χ_4	3	$\lambda + \lambda^2 + \lambda^4$	$\lambda^3 + \lambda^6 + \lambda^5$	0	0
χ_5	3	$\lambda^3 + \lambda^6 + \lambda^5$	$\lambda + \lambda^2 + \lambda^4$	0	0
χ_{reg}	21	0	0	0	0

□

Problem 4.4.

Proof. 1. The relation that $x^5 = 1$, tells us that x is diagonalizable with eigenvalues which are a part of the 5th roots of unity.

2. Since x is diagonalizable with eigenvalues of 5th roots of unity, λ_1, λ_2 . x^{-1} would have eigenvalues $\bar{\lambda}_1$ and $\bar{\lambda}_2$. Since the trace is conjugacy invariant, it means $\chi(x) = \lambda_1 + \lambda_2 = \bar{\lambda}_1 + \bar{\lambda}_2 = \chi(x^{-1}) \implies \chi(x) = \lambda_1 + \lambda_2 = \bar{\lambda}_1 + \bar{\lambda}_2 = \chi(x^{-1}) = \overline{\chi(x)} \implies \chi(x) \in \mathbb{R}$, this is only possible if $Im(\lambda_2) = -Im(\lambda_1)$. Thus $\sin(2\pi k_1/5) = \sin(-2\pi k_2/5)$, thus $k_2 = k_1^{-1}$ in μ_5 . Thus $\chi(x) = 2\cos(2k\pi/5)$ but $k = 1, 2$, because if $k > 2$, then you would get the same traces / eigenvalues.

3.

	e	x	x^2	y
χ_1	1	1	1	1
χ_2	1	1	1	-1
χ_3	2	$2\cos(\frac{2\pi}{5})$	$2\cos(4\pi/5)$	0
χ_4	2	$2\cos(4\pi/5)$	$2\cos(\frac{2\pi}{5})$	0
χ_{reg}	10	0	0	0

4. $\chi_A, \chi_B, \chi_C, \chi_D$ be some 1 dimensions of C_5 . Let $\omega = e^{2\pi i/5}$

	e	x	x^2	x^3	x^4
χ_A	1	ω	ω^2	ω^3	ω^4
χ_B	1	ω^4	ω^3	ω^2	ω^1
χ_C	1	ω^2	ω	ω^4	ω^3
χ_D	1	ω^3	ω^4	ω	ω^2

$$\chi_3|_{C_5} = \chi_A + \chi_B \text{ and } \chi_4|_{C_5} = \chi_C + \chi_D.$$

□

Problem 4.5.

Proof. $G = \mu_5 \rtimes \mu_4 = \langle x, y | x^5 = y^4 = e, yxy^{-1} = x^2 \rangle$. $|G| \leq 20$. $\mu_4 \rightarrow \text{Aut}(\mu_5) = \mu_4$, since every element has order 4 so sending $x \rightarrow x^2$ by conjugation is a valid automorphism of μ_5 . Thus $|G| = 20$.

$yxy^{-1} = x^2 \implies yx^2y^{-1} = x^4 \implies yxy^{-1} = x^3$ since conjugation is a homomorphism. Thus $[x] = x, x^2, x^3, x^4$.

$xyx^{-1} = xyx^{-1}y^{-1}y = x^4y \implies xx^4yx^{-1} = x^4xyx^{-1} = x^3y \implies xx^3yx^{-1} = x^3x^4y = x^2y \implies xx^2yx^{-1} = xy$. Thus $[y] = y, xy, x^2y, x^3y, x^4y$. Since conjugation is a homomorphism, $[y^2] = y^2, xy^2, x^3y^2, x^4y^2$ and similarly for $[y^3]$.

Thus the conjugacy classes are $[e], [x], [y], [y^2], [y^3]$

The dimensions of the irreducibles can be of sizes 1, 2, 4, 5, 10 since they must divide 20. The number of irreducibles is 5, number of conjugacy classes. The sum of the squares of the irreducibles is

20 and there is 1 trivial representation. Thus now the possible dimensions can be 1, 2, 4. However the only combination which works is 1 + 1 + 1 + 1 + 4.

μ_5 is a normal subgroup such that $G/\mu_5 = \mu_4$ which is abelian. Moreover, $[x, y] = xyx^{-1}y^{-1} = x^4 \in \mu_5$ moreover x^4 generates μ_5 . Thus $[G, G'] \subseteq \mu_5$ and $\mu_5 \subseteq [G, G']$. Thus $\mu_5 = [G, G']$. The 1 dimensional irreducibles of μ_4 send $y \rightarrow \pm 1, \pm i$. Thus the 1 dimensional irreducibles of G , send $x \rightarrow 1$ and $y \rightarrow \pm 1, \pm i$.

Let V_1, V_2, V_3, V_4 be the 1 dimension representation of G . Let V_5 be the 4 irreducible dimensional representation. Furthermore since $\chi_{reg} = \sum_{i=1}^4 \dim V_i * \chi_{V_i}$. But $\chi_{reg}(e) = 20$ and $\chi_{reg}(g) = 0 \forall g \in G \ni g \neq e$. $\chi_{V_5} = \frac{1}{4}(\chi_{reg} - \sum_{i=1}^4 \dim V_i * \chi_{V_i})$. Use this to find the characters of the 4 dimensional irreducibles, the calculated values are below.

	e	x	y	y^2	y^3
χ_1	1	1	1	1	1
χ_2	1	1	-1	1	-1
χ_3	1	1	i	-1	-i
χ_4	1	1	-i	-1	i
χ_5	4	-1	0	0	0
χ_{reg}	20	0	0	0	0

□

Problem 4.6.

Proof. To turn the matrix A made from the entries of the character table unitary you need to multiply each entry by the square root of the size of the conjugacy class divided by the size of the group.

Let the number of conjugacy classes be n . Let $x_1, \dots, x_n$ be the elements in distinct conjugacy classes.

$$B \text{ the transformed matrix. Thus } B = \frac{1}{|G|} \begin{bmatrix} \sqrt{|[x_1]|} \chi_1(x_1) & \dots & \sqrt{|[x_n]|} \chi_1(x_n) \\ \vdots & \ddots & \vdots \\ \sqrt{|[x_1]|} \chi_n(x_1) & \dots & \sqrt{|[x_n]|} \chi_n(x_n) \end{bmatrix}$$

Let χ_i and χ_j be the characters of 2 representations. $\sum_{k=1}^n \sqrt{|[x_k]|} \chi_i(x_k) \sqrt{|[x_k]|} \chi_j(x_k) = \sum_{k=1}^n |[x_k]| \chi_i(x_k) \chi_j(x_k) = \sum_{k=1}^n \sum_{g \in [x_k]} \chi_i(g) \chi_j(g)$, since characters are class functions. $\sum_{k=1}^n \sum_{g \in [x_k]} \chi_i(g) \chi_j(g) = \sum_{g \in G} \chi_i(g) \chi_j(g) = |G| \langle \chi_i, \chi_j \rangle$.

Thus $B B^T = \frac{1}{|G|} [|G| \langle \chi_i, \chi_j \rangle]_{i,j} = I$, because traces of irreducibles form an orthonormal basis and thus are orthogonal. Thus B is unitary.

Since B is unitary so is, $\overline{B^T}$ and B^T , which tells us the $\langle c_i, c_j \rangle = \delta(i, j)$ where c_i and c_j are columns of B under the standard hermitian dot product □

Problem 4.7.

Proof. Let $\pi : G \rightarrow G/N$. Let $\rho : G/N \rightarrow V$ be a irreducible representation of G/N . Define the representation, $\phi = \rho \circ \pi$ Suppose $W \leq V$ is a G -invariant subspace. Then $\forall g \in G, \phi(g)W \leq W \implies \rho(\pi(g))W \leq W$ thus W is a G/N invariant subspace. Since V is a irreducible representation of G/N , the only G/N invariant subspaces are 0 and V . Thus $W = 0$ or $W = V$. Thus V doesn't have any proper G -invariant subspaces and thus V is irreducible wrt G .

χ_G is the character of G in the representation V . χ_G factors through $\chi_{G/N}$ by 1st semester pre-homework and $\ker \chi_{G/N} \leq \ker \chi_G$. Thus $\chi_G = \chi_{G/N} \circ \pi$. $\langle \chi_G, \chi_G \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_G(g) \overline{\chi_G(g)} = \frac{1}{|G|} \sum_{g \in G} \chi_{G/N}(\pi(g)) \overline{\chi_{G/N}(\pi(g))}$. Since everything in the same coset will have the same character under $\chi_{G/N}$ and there are $|N|$ elements in every coset, $\langle \chi_G, \chi_G \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_{G/N}(\pi(g)) \overline{\chi_{G/N}(\pi(g))} =$

$\frac{1}{|G|} |N| \sum_{gN \in G/N} \chi_{G/N}(gN) \overline{\chi_{G/N}(gN)} = \frac{1}{|G|/|N|} \sum_{gN \in G/N} \chi_{G/N}(gN) \overline{\chi_{G/N}(gN)} = \langle \chi_{\frac{G}{N}}, \chi_{\frac{G}{N}} \rangle = 1$.
Thus by the Main Theorem, V is irreducible. $\square$

Problem 4.8.

Proof. $|G| = 1 + 3 + 6 + 6 + 8 = 24$. The sum of the squares of the irreducibles must equal 24. There are 5 conjugacy classes, but only 4 characters listed. So we need 1 more irreducible reps. The current sum of the squares of the dimension is $1 + 1 + 9 + 9 = 20$. So we need a irreducibles of dimension 2.

By a previous homework, we know $\chi_{\mathbb{C}^G}(g) = |G| * \delta_e(g)$ and $|\mathbb{C}^G| = 24$. We also know, $\chi_{\mathbb{C}^G} = \sum_i \dim V_i * \chi_{V_i}$ thus $\chi_{\mathbb{C}^G} = 1 * (\chi_1 + \chi_2) + 3 * (\chi_3 + \chi_4) + 2 * \chi_5 \implies \chi_5 = \frac{1}{2} [\chi_{\mathbb{C}^G} - 1 * (\chi_1 + \chi_2) - 3 * (\chi_3 + \chi_4)]$
The table below has the results.

	(1)	(3)	(6)	(6)	(8)
χ_1	1	1	1	1	1
χ_2	1	1	-1	-1	1
χ_3	3	-1	1	-1	0
χ_4	3	-1	-1	1	0
χ_5	2	2	0	0	-1
$\chi_{\mathbb{C}^G}$	24	0	0	0	0

$\square$

Problem 2.6. *Frobenius 2.6: In 2.5 we found $Z := Z(\mathbb{C}[G])$, the center of the group algebra. Note as a vector space it has $\dim |G/G|$. And in fact we have a vector space iso $\text{Hom}_{\text{sets}}(G/G, \mathbb{C})$ to Z , taking $f \rightarrow \sum_{g \in G} f(g)g$. (or we can put in the scaling factor $1/|G|$). take the function $\text{tr}_V * \dim(V)$ (I think the scaling is needed to get something nice). For V irreducible, what does it map to? Call it p_V . show p_V^2 (as an elements of this ring $\mathbb{C}[G]$) = p_V , the various p_V commute, and their sum is 1 – the identity in this ring, and the p_V together give a basis of Z . Prove the map $C \times \dots \times C \rightarrow Z$ that takes $(a_V) \times \sum a_V p_V$ (where the domain is the product of copies of C a ring in the dumb way, one copy for each irreducible V) is an iso.*

Proof. Let V, W be irreducibles G -reps where $V \neq W$. Let $P_v = \frac{\dim V}{|G|} \sum_{g \in G} \text{tr}_V g$. Since $\text{tr}_V \in \text{Hom}_{\text{sets}}(G/G, \mathbb{C})$, $P_v \in Z(\mathbb{C}[G])$ as proven in the previous homework. $\forall v \in V$, $P_v V = (\frac{\dim V}{|G|} \sum_{g \in G} \text{tr}_V g) V = \frac{\dim V}{|G|} \sum_{g \in G} \text{tr}_V g V \subseteq \frac{\dim V}{|G|} \sum_{g \in G} \text{tr}_V V = V$, by linearity so P_v preserves V . Moreover since $P_v \in Z(\mathbb{C}(G))$, P_v and g commute.

Thus $P_v \in \text{Hom}_G(V, V)$ and $P_v \in \text{Hom}_G(W, W)$. By Shur's Lemma and as proven previously, $P_v|_V = \lambda_v I$ and $P_v|_W = \lambda_w I$. Furthermore, $\dim V \lambda_v = \text{tr}_V(P_v|_V) = \frac{\dim V}{|G|} \sum_{g \in G} \text{tr}_V(g) \overline{\text{tr}_V(g)} = \dim V \langle \text{tr}_V, \text{tr}_V \rangle = \dim V$, by (Shur's Lemma) theorems. Thus $\lambda_v = 1$.

$\lambda_w \dim W = \text{tr}_W(P_v|_W) = \frac{\dim V}{|G|} \sum_{g \in G} \text{tr}_V(g) \overline{\text{tr}_W(g)} = \dim W \langle \text{tr}_V, \text{tr}_W \rangle = 0$, by orthogonality relations, (Shur's Lemma). Thus $\lambda_w = 0$. Thus P_v projects onto V .

Since $P_v|_V = I$ and $P_v|_W = 0$ where W is a irreducible representation not equal to V . Under any representation, $P_v = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$. Thu $P_v^2 = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} = P_v$.

All the P_v 's commute since $\text{tr}_V * \dim V$ are class functions and thus all P_v 's are in the center by the previous homework.

Let $\{V_i\}$ be the set of all irreducibles of G .

$$\begin{aligned}
\sum_{i=1} P_{V_i} &= \frac{1}{|G|} \sum_{i=1} \dim V_i \sum_{g \in G} \text{tr}_{V_i}(g)g \\
&= \frac{1}{|G|} \sum_{g \in G} g \sum_{i=1} \dim V_i \text{tr}_{V_i}(g) \\
&= \frac{1}{|G|} \sum_{g \in G} g \left[\sum_{i=1} \dim V_i \text{tr}_{V_i}(g) \right] \\
&= \frac{1}{|G|} \sum_{g \in G} \chi_{\text{reg}}(g) \\
&= \frac{1}{|G|} \chi_{\text{reg}}(e) \\
&= 1
\end{aligned}$$

By Frobenius 2, we know $\{\text{tr}_{V_i}\}$ is a basis for $\mathbb{C}^{G/G}$.

$\forall a_i \in \mathbb{C} \ni \sum_{i=1} a_i P_{V_i} = 0 = \frac{1}{|G|} \sum_{i=1} a_i \dim V_i \sum_{g \in G} \text{tr}_{V_i}(g)g = \frac{1}{|G|} \sum_{g \in G} \left[\sum_{i=1} \dim V_i a_i \text{tr}_{V_i}(g) \right] (g)g = 0$. Since G is a basis for $\mathbb{C}[G]$, it means, $\left[\sum_{i=1} \dim V_i a_i \text{tr}_{V_i}(g) \right] (g) = 0, \forall g \in G$. Thus $\sum_{i=1} \dim V_i a_i \text{tr}_{V_i} = 0$. Thus $\dim V_i a_i = 0 \implies a_i = 0$, since tr_{V_i} is a basis. Thus $\{P_{V_i}\}$ is linearly independent.

$\forall z \in Z(\mathbb{C}[G])$ thus $z = \sum_{g \in G} f(g)g$ for $f \in \text{Hom}_{\text{sets}}(G/G, \mathbb{C})$ by the previous homework. However since $\{\text{tr}_{V_i}\}$ is a basis for $\mathbb{C}^{G/G}$, $\{\dim V_i \text{tr}_{V_i}\}$ is a basis for $\mathbb{C}^{G/G}$. $f = \sum_{i=1} a_i \dim V_i \text{tr}_{V_i} = \sum_{i=1} a_i \dim V_i \text{tr}_{V_i}$. Thus $z = \sum_{g \in G} \sum_{i=1} a_i \dim V_i \text{tr}_{V_i}(g)g = \sum_{i=1} a_i \dim V_i \sum_{g \in G} \text{tr}_{V_i}(g)g = \sum_{i=1} a_i P_{V_i}$, thus $z \in \text{span}(P_{V_i})$.

Since P_{V_i} forms a basis, then $\forall (c_1, \dots, c_n) \in \mathbb{C} \times \dots \times \mathbb{C}$, then for all $z \in Z$, as proven above $z = \sum_{i=1}^n c_i P_{V_i}$. Thus the map is surjective.

Suppose $\forall (c_1, \dots, c_n), (b_1, \dots, b_n) \in \mathbb{C} \times \dots \times \mathbb{C}$, such that $\sum_{i=1} c_i P_{V_i} = \sum_{i=1} b_i P_{V_i} \implies \sum_{i=1} (c_i - b_i) P_{V_i} = 0$, since P_{V_i} is a basis, $c_i - b_i = 0 \implies c_i = b_i$. Thus $(c_1, \dots, c_n) = (b_1, \dots, b_n)$ $\square$